Detailed Error Report – Quantum SVM on Textual Disease Classification

The dataset consists of 1200 datapoints across two columns: "label" and "text". - 'label': Contains disease names (24 unique disease labels). - 'text': Natural language descriptions of the symptoms. Each disease has 50 distinct symptom descriptions, giving a balanced dataset for classification. List of Diseases: Psoriasis, Varicose Veins, Typhoid, Chicken Pox, Impetigo, Dengue, Fungal Infection, Common Cold, Pneumonia, Dimorphic Hemorrhoids, Arthritis, Acne, Bronchial Asthma, Hypertension, Migraine, Cervical Spondylosis, Jaundice, Malaria, Urinary Tract Infection, Allergy, Gastroesophageal Reflux Disease, Drug Reaction, Peptic Ulcer Disease, Diabetes.

■ Error Encountered: Qiskit encountered a problem while attempting to execute a Quantum Support Vector Machine (QSVM) classification using the `qiskit-machine-learning==0.6.0` version stack. Error Message (excerpt): TypeError: OneHotEncoder.__init__() got an unexpected keyword argument 'sparse_output'

■ Cause of the Error: The issue arises due to a version mismatch between `qiskit-machine-learning==0.6.0` and the installed version of `scikit-learn` (>=1.2). - In `scikit-learn==1.2` and newer, the `OneHotEncoder` class introduced a new keyword argument `sparse_output`. - However, older versions like `qiskit-machine-learning==0.6.0` expect the older `sparse` parameter instead. Hence, when `sparse_output` is passed to the older constructor, it raises a `TypeError`.

■■ Recommended Fix: Downgrade `scikit-learn` to a version compatible with `qiskit-machine-learning==0.6.0`: pip install scikit-learn==1.1.3 This version still uses the `sparse` keyword and will be fully compatible with Qiskit ML 0.6.0. ■ Optional: Freeze compatible versions in a `requirements.txt`: qiskit==0.44.1 qiskit-aer==0.13.0 qiskit-terra==0.25.1 qiskit-machine-learning==0.6.0 scikit-learn==1.1.3 pandas==1.5.3 numpy==1.23.5

■ Notes: - Always ensure all library versions are compatible when working with legacy Qiskit versions. - Consider using a virtual environment to isolate dependencies: python3.10 -m venv qml_env source qml_env/bin/activate # On Windows: .\qml_env\Scripts\activate pip install -r requirements.txt - If you're working with natural language inputs, preprocessing steps such as negation handling and TF-IDF encoding must be handled manually before passing data to QSVM, which only supports numeric inputs.